AARYAN SHANBHAG

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SUMMARY

Engineering Systems Design student at SUTD specialising in Business Analytics, with hands-on experience in **time-series forecasting**, **inventory performance analysis**, and operational data modelling to support supply chain decision-making. Proficient in **Python**, **Excel**, and **SQL** across forecasting, KPI monitoring, and decision-support reporting.

EDUCATION

Singapore University of Technology and Design
B. Eng. in Engineering Systems Design *Sept 2024 – Present*
Tentative Specialisation: Business Analytics and Operations Research

Global Indian Int'l School, Punggol
International Baccalaureate (IB) Diploma *June 2020 – May 2022*

COURSEWORK

40.011 – Data & Business Analytics (SBSTransit Industry Project)

- **Built predictive maintenance pipeline** for SBSTransit railway track degradation using **segmented regression** (Python Statsmodels & Matplotlib).
- Automated **multi-sheet preprocessing** and **feature engineering** using **Excel Macros & VBA.**
- Designed decision-support system predicting **time-to-failure**, producing structured performance reports tracking key **operational indicators** and highlighting variances for decision-making.

GEO ITS x SUTD – Overseas Field Study (Surabaya, Indonesia)

- Forecasted **fuel consumption** and **revenue** for Surabaya's public bus operator using **Holt–Winters.**
- **Built 3-scenario forecast** (optimistic/moderate/pessimistic) with 95% confidence intervals
- Conducted **Benefit-Cost Analysis** (BCA = 0.93) to evaluate fleet transition viability and propose rebalancing recommendations to **optimise resource allocation.**
- Modelled **time-series data** in Python/Jupyter and Excel; evaluated via **MAPE & RMSE**.
- Conducted field observations to link operational inefficiencies with data patterns.

02.137DH – Digital Humanities

- Built **end-to-end NLP pipeline** in Python/Jupyter using **spaCy dependency parsing** to extract 191 labelled verb-instances from unstructured news text into structured **pandas DataFrames.**
- **Engineered custom rule-based extractors** for grammatical voice, epistemic hedging, actor classification and lexical framing, operationalizing Critical Discourse Analysis theory as computable metrics.
- Applied **chi-square significance testing (scipy.stats)** and produced visualisations in Matplotlib, Seaborn to compare and quantify linguistic framing asymmetries across news outlets.

40.002 – Optimisation

- Developed **MILP-based ER patient triage model** for a hospital, optimising admission decisions (Discharge/ICU/Ward) with capacity and routing constraints using **Julia | JuMP | HiGHS.**

TECHNOLOGIES	THEORY
Excel – Pivot Tables, Macros, VBA, Analysis ToolPak **SQL –** Table Join, Excel-Python integration **Python –** pandas, Matplotlib, statsmodels, sk-learn, SciPy, spaCy, Seaborn **R –** tidyverse (dplyr, ggplot2) **Julia –** JuMP, HiGHS **Tableau, PowerBI –** DAX	**Financial Analysis and Forecasting:** Cash Flows | linear/logistic regression | NPV/IRR metrics Holt-Winters (exp. smoothing) | DuPont decomposition **Statistical Inference and Hypothesis Testing:** Bayesian inference | Confidence Intervals | chi 2 & t-tests ANOVA | bootstrap resampling | non-parametric tests **Optimisation –** LP/MILP | Simplex | Branch-and-Bound Dynamic Programming | Primal-Dual & Lagrangian duality